From Mass Ratios to Anomalous Magnetic Moments: Why the Koide Structure Predicts g-2 Scaling in T0 Theory

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Abstract

The Koide formula $Q = (m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$ is an empirical relation among charged lepton masses, confirmed experimentally to 0.001%, which the Standard Model cannot explain. In T0 theory, it emerges from the exponent structure $p = 3/2, 1, 2/3$. The lepton exponents form a geometric sequence $p_{n+1} = (2/3)\, p_n$ with ratio $q = 2/3$. The step $\Delta p_{\mu\to\tau} = 1/3$ follows from $p_\mu \cdot (1 - 2/3) = 1/3$, not from $1/D$ — the same step appears in the g-2 hierarchy.

We show that the **same** $1/3$-step generates the $g-2$ anomaly hierarchy $\Delta a_i = 4\pi/f^{q_i}$ with exponents $q = 5/3, 4/3$ — again separated by $1/3$. The consequence is a direct, parameter-free bridge from the experimentally known mass ratio $m_\tau/m_\mu = 16.817$ to the g-2 ratio prediction $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = (144/125) \cdot m_\tau/m_\mu \approx 19.4$. This leads to $a_\tau \approx 1.281 \times 10^{-3}$ — testable at Belle II and qualitatively different from the SM prediction of $\approx 1.18 \times 10^{-3}$.

.1 Introduction: The Koide Mystery

The charged lepton masses span three orders of magnitude:

$$m_e = 0.511 \text{ MeV}, \quad m_\mu = 105.658 \text{ MeV}, \quad m_\tau = 1776.86 \text{ MeV} \tag{1}$$

The Standard Model treats these as free parameters — measured, not explained. Yet they satisfy the Koide relation [1] with extraordinary precision:

$$\boxed{Q = \frac{m_e + m_\mu + m_\tau}{\left(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau}\right)^2} = 0.666661 \approx \frac{2}{3}} \tag{2}$$

confirmed to $\Delta Q/Q = 0.001\%$. This precision demands explanation, yet the SM offers none. It is as if the masses know about each other — they satisfy a collective relation that no accident could produce.

.2 T0 Explanation of Koide: The Exponent Structure

In T0 theory, all lepton masses derive from a single parameter $\xi = 4/30{,}000$ via the Yukawa structure [2]:

$$m_i = r_i \cdot \xi^{p_i} \cdot v, \quad v = 246 \text{ GeV (Higgs VEV)} \tag{3}$$

with rational prefactors r_i and exponents p_i:

Lepton	r_i	p_i	m_{T0} (MeV)	m_{exp} (MeV)	$\Delta\%$
e	$4/3$	$3/2$	0.505	0.511	-1.2%
μ	$16/5$	1	104.96	105.66	-0.7%
τ	$25/9$	$2/3$	1783.4	1776.9	$+0.4\%$

Table 1: T0 lepton masses from a single parameter ξ

The exponents form a sequence:

$$p_e = \frac{3}{2}, \quad p_\mu = 1, \quad p_\tau = \frac{2}{3} \tag{4}$$

The step between muon and tau is $\Delta p_{\mu\to\tau} = 1/3$, while the step from electron to muon is $\Delta p_{e\to\mu} = 1/2$. The $1/3$ structure of the torus geometry manifests itself exactly in the μ-τ transition — and, as we will show, in the entire g-2 hierarchy.

This step of $1/3$ is not arbitrary. The lepton exponents form a geometric sequence with ratio $q = 2/3$: $(p_e, p_\mu, p_\tau) = (3/2, 1, 2/3)$. The step $\Delta p_{\mu\to\tau} = p_\mu \cdot (1-q) = 1 \cdot 1/3 = 1/3$ follows from this ratio, not directly from $1/D$. The different step $1/2$ for the electron ($p_e \cdot (1-q) = 3/2 \cdot 1/3 = 1/2$) is a consequence of the same geometric sequence.

The Koide relation then follows naturally: the specific combination of rational prefactors $(4/3, 16/5, 25/9)$ and the geometric progression $\xi^{3/2}, \xi^1, \xi^{2/3}$ conspire to produce $Q \approx 2/3$. What Koide discovered empirically, T0 derives from the torus geometry [3].

.3 From Masses to $g-2$: The Same Step Size

The $g-2$ Formulas in T0

The anomalous magnetic moments in T0 theory are [4]:

$$a_e = \frac{S_3}{f \cdot k_{\text{eff}}}, \quad S_3 = 2\pi^2, \quad f = 1/\xi = 7500 \tag{5}$$

$$a_\mu = a_e + \frac{4\pi}{f^{5/3}} \tag{6}$$

$$a_\tau = a_e + \frac{4\pi}{f^{4/3}} \tag{7}$$

The lepton-specific corrections Δa_i above the electron baseline have the structure:

$$\Delta a_i = \frac{4\pi}{f^{q_i}}, \quad q_\mu = \frac{5}{3}, \quad q_\tau = \frac{4}{3} \tag{8}$$

The Key Observation: Same $1/3$ Step

The $g-2$ exponents:

$$q_\mu = \frac{5}{3}, \quad q_\tau = \frac{4}{3}, \quad \Delta q = \frac{1}{3} \tag{9}$$

The mass exponents (Eq. 4):

$$p_\mu = 1, \quad p_\tau = \frac{2}{3}, \quad \Delta p = \frac{1}{3} \tag{10}$$

$$\boxed{\Delta p_{\mu\to\tau} = \Delta q_{\mu\to\tau} = \frac{1}{3} = \frac{1}{D}} \tag{11}$$

This is the central result: **The step between the heavier lepton generations ($\mu \to \tau$) is $1/3$ in both the mass hierarchy and the g-2 hierarchy.** This is not a fit — it follows from the geometric sequence $p_{n+1} = (2/3)\, p_n$, where $p_\mu \cdot (1 - 2/3) = 1/3$. The bridge formula uses precisely this μ-τ transition.

Why This Must Be So

In T0 theory, both mass and anomalous moment arise from how a lepton couples to the torus winding structure:

- The **mass** measures the coupling strength to the vacuum (ξ-field), governed by ξ^{p_i}

- The **anomalous moment** measures the deviation from the Dirac value due to the geometric environment, governed by f^{-q_i}

Both are aspects of the same winding mode. The step $1/3$ between the heavier generations follows from the geometric sequence $p_{n+1} = (2/3)\, p_n$ — the ratio $q = 2/3$ is numerically identical to the Koide value $Q = 2/3$.

.4 The Bridge: Mass Ratios → $g-2$ Ratios

The $f^{1/3}$ Factor

From the mass formulas:

$$\frac{m_\tau}{m_\mu} = \frac{r_\tau}{r_\mu} \cdot \xi^{p_\tau - p_\mu} = \frac{25/9}{16/5} \cdot \xi^{-1/3} = \frac{125}{144} \cdot f^{1/3} \tag{12}$$

From the $g-2$ formulas:

$$\frac{\Delta a(\tau{-}e)}{\Delta a(\mu{-}e)} = \frac{4\pi/f^{4/3}}{4\pi/f^{5/3}} = f^{1/3} \tag{13}$$

The factor $f^{1/3}$ appears in both. Combining Eqs. 12 and 13:

$$\boxed{\frac{\Delta a(\tau{-}e)}{\Delta a(\mu{-}e)} = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu}} \tag{14}$$

This is the bridge: **The $g-2$ ratio is determined by the mass ratio**, up to the known rational factor $144/125 = 1.152$. No free parameters. No α. No k_{eff}. No K_{frak}. Everything cancels except the mass ratio and the rational prefactors.

Numerical Verification

Using the experimentally known mass ratio:

$$\frac{m_\tau}{m_\mu} = \frac{1776.86}{105.658} = 16.817 \tag{15}$$

$$\frac{144}{125} \times 16.817 = 19.373 \tag{16}$$

Directly from $f^{1/3}$:

$$f^{1/3} = 7500^{1/3} = 19.574 \tag{17}$$

The 1.0% difference between 19.373 and 19.574 reflects the small deviation of T0 masses from experiment (Table 1). Both values are far from the SM ratio (see Section .5).

Tau Prediction

The prediction for a_τ follows in one step from the bridge formula and three experimental numbers:

1. **Measured difference:** $\Delta a(\mu{-}e) = a_\mu - a_e = 6.269 \times 10^{-6}$
2. **Mass ratio** (experimental, high-precision): $m_\tau/m_\mu = 16.817$
3. **Bridge formula** (Eq. 14): $\Delta a(\tau{-}e) = \frac{144}{125} \cdot \frac{m_\tau}{m_\mu} \cdot \Delta a(\mu{-}e)$

Substituting:

$$\Delta a(\tau\text{–}e) = 1.152 \times 16.817 \times 6.269 \times 10^{-6} = 1.214 \times 10^{-4} \tag{18}$$

$$a_\tau = a_e + \Delta a(\tau\text{–}e) = \boxed{1.281 \times 10^{-3}} \tag{19}$$

That is all. No α, no k_{eff}, no K_{frak}, no perturbation theory, no 12,672 Feynman diagrams. Only two measured g-2 values, a mass ratio, and the rational factor $144/125$.

Why so simple? The entire complexity of T0 theory cancels out. From $\Delta a_i = 4\pi/f^{q_i}$ follows $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = f^{1/3}$, and from the masses follows $m_\tau/m_\mu = (125/144)\cdot f^{1/3}$. Combining yields the bridge — independent of the exact value of f, ξ, or any other T0 parameter.

Consistency check: The SM calculates $a_\tau^{\text{SM}} = 1.177 \times 10^{-3}$, corresponding to a ratio $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = 2.80$. T0 predicts 19.4 — a factor of 7, clearly testable.

T0-internal path: Computing all quantities purely from T0 geometry (Doc 018) yields $a_\tau^{\text{T0}} = 1.245 \times 10^{-3}$. The difference ($1.245$ vs. 1.281) arises because $\Delta a(\mu\text{–}e)_{\text{exp}}$ contains the full SM corrections, while the T0-internal difference $4\pi/f^{5/3}$ does not.

Method	a_τ ($\times 10^{-3}$)	Remark
Geometric projection	1.282	$f^{1/3} \times \Delta a(\mu\text{–}e)_{\text{exp}}$
T0 internal	1.245	All quantities from T0
T0 range	1.25–1.28	
SM	1.177	Ratio 2.80

Table 2: Tau predictions: T0 vs. SM

.5 Comparison with the Standard Model

Mass Ratios: SM Has No Prediction

The SM treats all lepton masses as free parameters. It cannot predict m_τ/m_μ, cannot explain Koide, and has no mass hierarchy formula. The experimental ratios $m_\mu/m_e = 206.77$, $m_\tau/m_\mu = 16.82$ are inputs, not outputs.

T0 derives all three masses from ξ, explains Koide, and predicts the ratios to $\sim 1\%$.

$g - 2$ Ratios: Qualitatively Different Predictions

The SM calculates each lepton's $g - 2$ separately using perturbative Feynman diagrams (up to 12,672 at fifth order for the electron). No scaling relation between leptons is built in — the ratio of anomalies simply falls out of the calculation.

Ratio	T0	SM	Factor
$\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e)$	19.57	2.80	7.0×
a_τ prediction	1.281×10^{-3}	1.177×10^{-3}	+9%

Table 3: T0 vs SM: $g - 2$ ratio predictions for the tau lepton

This is not a subtle 0.1% difference. The T0 $g - 2$ ratio is **seven times larger** than the SM ratio. The tau anomaly differs by 9%. Belle II will decide.

The Fair Comparison: Equal Starting Conditions

A common objection is that SM calculations of a_e achieve 10^{-12} precision, far beyond T0. This comparison is however misleading:

- **SM circularity:** The SM extracts α from the a_e measurement itself. Calculating a_e with this α is circular — a fit, not a prediction.
- **T0 derives α:** From the single free parameter $\xi = 4/30000$ follows $\alpha = \xi \cdot E_0^2$ with the characteristic energy $E_0 = 7.398$ MeV, yielding $1/\alpha_{\text{T0}} = 137.0353$ — only 0.0005% from the experimental value $1/\alpha_{\text{exp}} = 137.0360$.
- **SM with T0-α:** When the SM uses α_{T0} instead of α_{exp}, the results are practically identical — the 0.0005% difference in α produces only $\sim 5 \times 10^{-9}$ shift in a_μ, well within experimental uncertainty. The SM loses no precision for electron and muon when using T0-α.
- **The decisive difference:** The SM requires α as measured input. T0 derives it from ξ. The SM has *no* α without measurement — T0 has one without any measurement, and it agrees to 5×10^{-6}.

Furthermore, T0 computes the anomalous moment directly from geometry: $a_e = S_3/(f \cdot k_{\text{eff}}) = 1.1598 \times 10^{-3}$, only 0.014% from experiment — *without* the detour through α and perturbative expansion.

Where the theories truly diverge: Not at electron or muon, but at the **tau**. T0 predicts $a_\tau \approx 1.281 \times 10^{-3}$ (from the bridge formula), the SM predicts $a_\tau \approx 1.177 \times 10^{-3}$. That is a 9% difference — testable at Belle II.

The ratio prediction (Eq. 14) sidesteps the α debate entirely: it uses experimental a_e, a_μ, and m_τ/m_μ. The only theoretical input is the rational factor $144/125$.

Background: Why 12,672 Diagrams?

The SM calculation of $g-2$ follows a clear algorithm. All of QED derives from a single line — the Lagrangian density:

$$\mathscr{L} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \tag{20}$$

This contains exactly **three building blocks** (vertices): A fermion couples to a photon ($\bar{\psi}\gamma^\mu A_\mu \psi$), a fermion propagates freely, a photon propagates freely. From these three building blocks, all Feynman diagrams are assembled like Lego bricks:

- Each vertex couples exactly one photon to a fermion line and contributes a factor $\sqrt{\alpha}$
- Order n = all diagrams with n vertices, contributing $\alpha^{n/2}$
- One draws *all topologically possible* diagrams at a given order
- Each diagram is translated into an integral using the Feynman rules
- All diagrams are summed

The *all topologically possible* is where the computational cost lies: At order 1 there is 1 diagram (Schwinger 1948), at order 2 there are 7, at order 3 already 72, at order 4 there are 891, and at order 5 finally 12,672 — each one a multi-dimensional integral that must be evaluated.

How does one know the Lagrangian is correct? It follows from three symmetry principles: Lorentz invariance (relativity), local $U(1)$ gauge symmetry (forces the existence of the photon), and renormalizability. One postulates these symmetries and derives the unique possible theory. Then one calculates and compares with experiment — and it agrees to 12 decimal places.

But *nobody explains why these symmetries hold*. Why $U(1)$? Why is $\alpha = 1/137$ and not $1/200$? The SM says: that is how it is, measure it.

Order	Diagrams	Contribution ∼	Effort
1	1	$\alpha/2\pi$	Analytical (Schwinger 1948)
2	7	α^2	Analytical
3	72	α^3	Partly numerical
4	891	α^4	Monte Carlo, decades of work
5	12,672	α^5	Automatic code generation

Table 4: Growth of Feynman diagrams by order in QED

T0 provides the why: The Lagrangian is not the foundation — the torus geometry is. The $U(1)$ symmetry is a consequence of the geometric structure, not an axiom. The fine-structure constant follows from $\alpha = \xi \cdot E_0^2$ and the anomalous moments directly from geometry ($a_e = S_3/(f \cdot k_{\text{eff}})$) without perturbative expansion. The SM has a perfect recipe, but no explanation for why the ingredients work together. T0 provides the why — and arrives with a single formula at results that require the SM 12,672 diagrams and decades of computing time.

The fair comparison is therefore not accuracy versus accuracy, but **explanatory depth versus computational precision** — complementary strengths.

.6 The Argument Chain

The logic is:

1. The Koide formula holds experimentally to 0.001%. This is an unexplained empirical fact.
2. T0 explains Koide through the geometric sequence $p = 3/2, 1, 2/3$ with ratio $q = 2/3$; the step $1/3$ in the μ-τ transition follows from $p_\mu \cdot (1-q) = 1/3$.
3. The **same** $1/3$ step appears in the $g-2$ exponents $q = 5/3, 4/3$.
4. This is not a coincidence — it is the same torus winding structure manifesting in two observables.
5. Therefore: the $g-2$ ratio $\Delta a(\tau\text{–}e)/\Delta a(\mu\text{–}e) = f^{1/3}$ is as well-founded as the mass hierarchy. It rests on the same geometry.
6. Combining with bridge formula: $a_\tau \approx 1.281 \times 10^{-3}$.

If one accepts that the Koide relation is not accidental — and its 0.001% precision makes accident implausible — then one must also accept that the $g-2$ ratio prediction has the same geometric foundation. The predictions are linked. They cannot be separated.

.7 Belle II: The Decisive Test

Belle II at KEK expects sensitivity of $\sim 10^{-7}$ for a_τ. The two predictions are far apart:

Theory	a_τ ($\times 10^{-3}$)
SM (perturbative)	1.177
T0 (bridge formula, Eq. 19)	1.281

Table 5: Tau anomaly predictions — Belle II will distinguish

The difference is 9% — not a subtle effect, but a qualitative distinction. Three outcomes are possible:

- $a_\tau \approx 1.18 \times 10^{-3}$: SM confirmed, T0 falsified.
- $a_\tau \approx 1.28 \times 10^{-3}$: T0 confirmed. The mass-g-2 bridge is real, and the Koide structure extends to anomalous moments.
- a_τ outside both ranges: both theories require revision.

.8 Conclusion

The Koide formula is not an isolated curiosity. In T0 theory, it is one manifestation of a deeper geometric structure: the $1/3$-step between lepton generations, originating from the three-dimensional torus topology. This same structure predicts the $g-2$ anomaly hierarchy, connecting two seemingly unrelated observables through a single geometric principle.

The mass ratio $m_\tau/m_\mu = 16.82$ — experimentally known to high precision — directly determines via the bridge formula the g-2 ratio $\Delta a(\tau{-}e)/\Delta a(\mu{-}e) \approx 19.4$, leading to $a_\tau \approx 1.281 \times 10^{-3}$. This prediction is α-independent and qualitatively different from the SM value of 1.18×10^{-3}.

The Standard Model cannot make this connection. It has no explanation for Koide, no scaling relation between generations, and no bridge from masses to anomalous moments. It treats each observable as independent. T0 treats them as two faces of one geometry.

Belle II will decide.

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Addendum P1: Clarification of Precision Values (7 September 2026)

See addendum P1 in Doc. 116: the values $\Delta Q < 0.00003\,\%$ and $0.001\,\%$ are both correct and describe different contexts (internal calculation vs. external presentation with measurement uncertainty of the input masses).